



Regular Article

Unraveling the link of mental health problems with religiosity, internet addiction, and sleep quality among medical students in Bangladesh

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ABSTRACT

Background: The purpose of the study was to examine the association of religiosity, internet addiction, and sleep quality with mental health problems among medical students in Bangladesh.

Methods: This cross-sectional study involved 500 medical students from four medical universities, selected through a multistage random sampling process. Mental health problems, internet addiction, and sleep quality were assessed using the Bangla-adapted questionnaires. The chi-square test and binary logistic regression analysis were used to determine the prevalence and associated factors of mental health concerns related to religiosity.

Results: A total of 54.2% of students suffered from mental health issues. The significant associated factors of mental health problems were gender, marital status, smoking status, sleep quality, and internet addiction. The findings revealed that non-religious students experienced 2.63 times more mental health-related issues than religious students (aOR = 2.63, 95% CI: 1.61–4.30). Participants with internet addiction and poor sleep quality had 3.1 times (aOR = 3.10, 95% CI: 2.05–4.71) and 4.36 times (aOR = 4.36, 95% CI: 2.21–8.63), respectively, higher mental health problems compared to those without internet addiction and good sleep quality.

Conclusion: Religiosity, internet addiction, and poor sleep quality have significantly correlated with the mental health of medical students. So, these findings may aid medical authorities and mental health practitioners in boosting the mental health of medical students.

1. What we already know

1. Mental health was significantly influenced by the sleep quality and internet addiction.
2. There was a significant negative correlation between mental health problems and religiosity.

2. What this article adds

1. This study confirmed that religiosity, internet addiction, and sleep quality are combinedly associated with mental health providing a more specific understanding in a specific cultural context.
2. Region specific data help the understanding the medical students' mental health, which may differ from other countries such as Western countries and also Asian countries.

3. Evidence from lower-middle-income countries like Bangladesh can be strengthen the global understanding of medical students' mental health.

3. Introduction

Mental health problems are now recognized as an increasing global concern, affecting quality of life and work productivity (Al-Mamun et al., 2024). Poor mental health significantly decreases quality of life, impairs daily activities, and may result in further physical difficulties. Students are increasingly recognized as being more vulnerable to mental health problems than the general population, due to facing challenges and requiring autonomous decision-making over their lives and studies. In particular, medical students are more prone to a wide range of mental health issues compared to their peers in other academic disciplines

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(Wilkinson, 2023), because they have always been one of the toughest courses to study (Willcock et al., 2004). A study in Bangladesh had found that almost half of the medical students had a poor mental health status (Abbasy et al., 2022). However, many factors influence the mental health of medical students. Many studies have found that age (Kerebih et al., 2017), gender (Kerebih et al., 2017; Rtbey et al., 2022), marital status (Kerebih et al., 2017), smoking a cigarette (Kerebih et al., 2017) were connected to mental health difficulties among the medical students. Other research also suggested that internet addiction (IA), smoking, and sleep quality influence mental health (Banna et al., 2023; Boksa, 2017; Sayeed et al., 2021; Scott et al., 2021).

In Bangladesh, religiosity can influence the mental health of medical students. Religiosity refers to people's varying tendencies to commit themselves to religious beliefs, principles, and activities (Leite et al., 2023). It serves as a protective factor for poor mental health (Malinakova et al., 2024; Yusoff et al., 2010). A recent study on medical students revealed that religion and the level of religious practice were significantly associated with mental health (Sandu et al., 2025). Religion is a socially and culturally identifying variable, and religiosity represents the level of personal beliefs, practice and involvement in religious values. Individuals may identify a specific affiliation (e.g., Muslim, Hindu, or other religion), but vary significantly in their level of practice and commitment to religious values. This distinction is especially important in the Bangladeshi context. Bangladesh is a mostly Muslim country, where religious identity is often culturally embedded and broadly acknowledged. Including both religion and religiosity as explanatory variables enables us to explore the associations between mental health issues and cultural identity (religious affiliations) and the degree of religious engagement (religiosity).

A study conducted on Bangladeshi medical students identified the risk factors for mental health during COVID-19 (Ashiq et al., 2023). Some Bangladeshi studies have been conducted on the association between internet addiction, sleep quality, and mental health. But these studies were conducted on a different population. For example, a recent study found that sleep quality was significantly associated with mental health problems such as anxiety, depression, and stress among medical students in Bangladesh (Muyeed et al., 2026). Internet addiction and poor sleep quality were significantly associated with mental health problems among university students in Bangladesh (Mondal et al., 2025). Variables such as internet addiction, sleep quality, and mental health issues were examined independently rather than in an integrated framework. There was a paucity in research examining the association between religiosity and mental health among Bangladeshis, both in the general population and among medical students. Bangladesh is a highly religious country, where religious practice and belief play a crucial role in its life (Devine et al., 2019). However, no published quantitative studies have examined religiosity and mental health from the perspective of Bangladesh. Therefore, a clear gap exists in that there is no comprehensive study that simultaneously examined the association of sleep quality, internet addiction, and religiosity with mental health issues among medical students in Bangladesh. To address this gap, the current study aims to explore the association between mental health, religiosity, internet addiction, and sleep quality among medical students in Bangladesh. Addressing this gap may provide a comprehensive, culture-specific understanding of mental health and its associated factors, as well as effective intervention techniques tailored to this population.

4. Methods

4.1. Study design, study settings

This study was cross-sectional, and data were collected from four medical colleges in Bangladesh. Four medical universities were Rajshahi Medical College (RMC), Mymensingh Medical College (MMC), Sir Salmullah Medical College (SSMC), and MAG Osmani Medical College

(MOMC). Students came from various parts of Bangladesh to study at these four medical colleges.

4.2. Sample size and sampling method

The sample size estimated by the equation: $n = \frac{Z^2 P(1-p)}{d^2}$, where, n = sample size; $z = 1.96$ (95% CI); p = prevalence estimate (0.335) as per a Bangladeshi study (Hasan et al., 2022); and $d = 0.05$. So, a total of 342 participants were required. However, we approached 560 students to get better outcomes. The data were collected through appointed research assistants. Firstly, We randomly selected four medical colleges (using a lottery method) from the existing government medical colleges in Bangladesh. Secondly, we randomly selected 8 halls (5 male and 3 female) from the four medical colleges. We selected 70 participants randomly from each hall. Consequently, we selected 560 (8×70) participants (210 females, 350 males). However, 52 participants (9 females, 43 males) did not give their consent to participate in this study; 8 students (1 female, 7 males) did not complete the questionnaires correctly. So, we excluded 60 participants. Thus, the final sample was 500. The sampling process was entirely random. The duration of the collection was August 2024 to January 2025.

4.3. Inclusion and exclusion criteria

The inclusion criteria were that the students were from a medical college in Bangladesh and were in the first to final years.

Exclusion criteria for this study were students who were not Bangladeshi citizens and students studying in private medical colleges. Students having serious physical (diagnosed by a doctor) and mental health problems (diagnosed by a psychiatrist).

4.4. Ethical consent

The ethical approval was taken from the Ethical Review Committee, Rajshahi Medical College, Rajshahi-6000, Bangladesh (Ref. RMC/ERC/2024/158). We also consider the guidelines provided by the 1964 Helsinki Declaration. We asked the participants for their consent to participate in this study. If they agreed, then we provided questionnaires. Then, we discussed the study's purpose and relevant questionnaires before collecting the final data. We did not provide any financial aid to participate in this study. All information was collected anonymously to consider the privacy and confidentiality of the participants.

4.5. Outcome variable

The outcome variable for this study was the mental health status of the participants. The Bangla-adapted General Health questionnaire measured the mental health of the medical students (Islam et al., 2024).

4.6. Explanatory variables

The explanatory variables for this study were age, gender (male and female), marital status (married and unmarried), smoking status (yes and no), sleep quality (good and poor), religion (Muslim and Non-Muslim), Internet addiction (absent and present), and religiosity (religious and non-religious).

4.7. Measuring tools

4.7.1. General Health Questionnaire (GHQ-12)

GHQ-12 was a very reliable self-reporting tool for assessing mental health conditions (Goldberg et al., 1997). This scale consists of 12 items on a 4-point Likert scale (0-3) and yields a total score of 0-36. The cutoff point differs for GHQ-12 across cultures and regions (Goldberg et al., 1998). The cutoff score considered 16 for this study, which was used

earlier in a Bangladeshi study, indicated good and poor mental health status (Mumenin et al., 2024). The Cronbach's alpha for this scale was 0.81, indicating good reliability.

4.7.2. Internet Addiction Test (IAT)

The IAT was a valid self-reporting tool used to assess internet addiction (Young, 1998). The Bangla-adapted IAT was used to measure internet addiction (Rezaul Karim & Nigar, 2014). This scale has 20 items with a Five-point Likert scale (0 to 5). The total score ranges from 20 to 100. A cutoff score of 50 was used for indicated internet addiction worldwide, and also in Bangladeshi culture (Rezaul Karim & Nigar, 2014; Young, 1998). The Cronbach alpha was 0.85, suggesting good internal consistency.

4.7.3. Pittsburgh Sleep Quality Index (PSQI)

The PSQI scale was a self-reporting scale used to screen sleep quality (Buysse et al., 1989). The Bangla-adapted PSQI scale was used to measure the sleep quality of medical students (Mondal et al., 2018). This scale has 19 items, organized into 7 components, and each component is weighted from 0 to 3 points. The total score ranged from 0 to 21 (Buysse et al., 1989). A score higher than 5 indicates poor sleep quality (Mondal et al., 2018). The Cronbach alpha was 0.78, suggesting satisfactory internal consistency.

4.7.4. Religiosity

In this present study, we measure religiosity by one question: "Do you practice your religion?". Here were two self-reporting options for response: (i) yes = considered as higher religious, and (ii) no = considered as lower religious or no (Berthold & Ruch, 2014). This single question was previously used among Bangladeshi students to measure religiosity (Mondal et al., 2024).

4.8. Statistical analysis

Firstly, we perform descriptive analysis to summarize the participants' characteristics. Secondly, the chi-square test was employed to find the prevalence of mental health status among medical students. Thirdly, the significant factors from the chi-square test were entered into the binary logistic regression model. We also checked the multicollinearity problem using the variance inflation factor (VIF); there are no multicollinearity problems if $0 \leq \text{VIF} \leq 5$ (Kim, 2019). The value of VIF satisfies no multicollinearity problem for this study. We also checked model fit for the binary logistic model by Pseudo- R^2 , the Akaike Information Criterion (AIC), and the Bayesian Information Criterion (BIC). We interpreted a higher pseudo R^2 and lower AIC and BIC values compared to other models, indicating superior goodness of fit for that model. A significant result was considered if the P -value was < 0.05 . All statistical tests were done by Stata 17.

5. Results

5.1. Background characteristics and mental health prevalence

All characteristics of the participants and the prevalence of mental health problems are shown in Table 1. The total number of respondents was 500, of which the majority were male (60.0%) and unmarried (92.6%). A significant portion of the respondents (89.2%) reported not smoking, and 14.4% reported poor sleep quality. Most participants were Muslim (91.0%) and self-identified as non-religious (77.8%). More than half (56.6%) were found to have internet addiction. Overall, 54.2% of the participants reported symptoms of mental health problems.

A significantly higher proportion of females (64.5%) reported mental health issues compared to males (47.3%) ($p < 0.001$). Mental health problems were also significantly more common among unmarried individuals (56.2%) compared to married individuals (29.7%) ($p < 0.001$). Notably, those who reported poor sleep quality had a higher prevalence

Table 1

Characteristics and the prevalence of mental health problems of the respondents.

Variables	Frequency (%)	Mental health problems		p-values
		No	Yes	
Age				0.291
≤19	117 (23.4)	52 (44.4)	65 (55.6)	
20-22	213 (42.6)	91 (42.7)	122 (57.3)	
≥23	170 (34.0)	86 (50.6)	84 (49.4)	
Gender				0.000
Male	300 (60.0)	158 (52.7)	142 (47.3)	
Female	200 (40.0)	71 (35.5)	129 (64.5)	
Marital status				0.000
Married	37 (7.4)	26 (70.3)	11 (29.7)	
Unmarried	463 (92.6)	203 (43.8)	263 (56.2)	
Body mass index				0.119
Underweight	62 (12.4)	23 (37.1)	39 (62.9)	
Normal Weight	385 (77.0)	186 (48.3)	199 (51.7)	
Overweight and Obese	53 (10.6)	20 (45.8)	33 (62.2)	
Family income (in taka)				0.373
≤20000	236 (47.2)	106 (44.9)	130 (55.1)	
20001-30000	96 (19.2)	50 (52.1)	46 (47.9)	
>30000	168 (33.6)	73 (43.5)	95 (56.5)	
Residence				0.435
Urban	163 (32.6)	76 (46.6)	87 (53.4)	
Rural	337 (67.4)	153 (45.4)	184 (54.6)	
Smoking status				0.007
No	446 (89.2)	195 (43.7)	251 (56.3)	
Yes	54 (10.8)	34 (62.9)	20 (37.1)	
Sleep quality				0.000
Good	72 (14.4)	58 (80.6)	14 (19.40)	
Poor	428 (86.6)	171 (39.9)	257 (60.1)	
Religion				0.288
Muslim	455 (91.0)	205 (45.0)	250 (55.0)	
Non-Muslim	45 (9.0)	24 (53.3)	21 (46.7)	
Internet addiction				0.000
Absent	217 (43.4)	142 (65.4)	75 (34.6)	
Present	283 (56.6)	87 (30.7)	196 (69.3)	
Religiosity				0.000
Religious	111 (22.2)	73 (65.8)	38 (34.2)	
Non-religious	389 (77.8)	156 (40.1)	233 (59.9)	
Total	500 (100)	229 (45.8)	271 (54.2)	

of mental health problems (60.1%) than those with good sleep quality (19.40%) ($p < 0.001$). Participants with IA had a significantly greater prevalence of mental health issues (69.3%) than those without addiction (34.6%) ($p < 0.001$). Non-religious participants were also more likely to experience mental health issues (59.9%) than their religious counterparts (34.2%) ($p < 0.001$).

5.2. Associated factors of mental health

A multivariate binary logistic regression model was developed based on the variables significantly linked with mental health problems in the chi-square analysis (Table 1). Results are presented in Table 2. Female students had 2.39 times (95% CI: 1.54-3.69) higher odds of experiencing mental health problems compared to male students. Unmarried individuals were 3.35 times (95% CI: 1.48-7.55) more likely to suffer from mental health issues than their married counterparts. Students who smoked had 2.08 times (95% CI: 1.07-4.05) higher odds of mental health problems than non-smokers. Participants with poor sleep quality had 4.36 times (95% CI: 2.21-8.63) higher odds of mental health problems than those with good sleep quality. Similarly, individuals with internet addiction were 3.10 times (95% CI: 2.05-4.71) more likely to experience mental health issues than those without internet addiction. Non-religious individuals had 2.63 times (95% CI: 1.61-4.30) higher odds of mental health problems than religious individuals.

6. Discussions

This study estimates the prevalence of mental health problems

Table 2
Associated factors of mental health.

Variables	Model 1	Model 2	Model 3	Model 4
	COR	AOR	AOR	AOR
Gender				
Male (ref.)				
Female	2.02 (1.39-2.92)	2.20 (1.51-3.21)	2.33 (1.52-3.57)	2.39 (1.54-3.69)
Marital status				
Married (ref.)				
Unmarried	3.03 (1.46-6.27)	3.64 (1.72-7.70)	3.14 (1.41-6.98)	3.35 (1.48-7.55)
Smoking status				
No (ref.)				
Yes	2.19 (1.22-3.92)		1.83 (0.96-3.52)	2.08 (1.07-4.05)
Sleep quality				
Good (ref.)				
Poor	6.23 (3.36-11.52)		4.42 (2.27-8.61)	4.36 (2.21-8.63)
Internet addiction				
Absent (ref.)				
Present	4.26 (2.93-6.21)		3.37 (2.24-5.08)	3.10 (2.05-4.71)
Religiosity				
Religious (ref.)				
Non-religious	2.87 (1.85-4.46)			2.63 (1.61-4.30)
Pseudo R ²		0.0393	0.1612	0.1835
AIC		668.48	590.46	577.10
BIC		681.13	615.75	606.60

Note: ‘COR, crude odds ratio, ‘AOR, adjusted odds ratio, ‘AIC, Akaike Information Criterion, ‘BIC, Bayesian Information Criterion, odds ratios in bold indicate statistically significant.

among medical students and examines their association with religiosity. The findings indicate that more than half of the participants are experiencing mental health problems. This prevalence is consistent with recent studies conducted among Bangladeshi medical students (Abbasy et al., 2022; Muyeed et al., 2026), as well as among different countries', such as Indians (Dwivedi et al., 2021), Pakistani (Babar et al., 2024), and Nepali (Kafle et al., 2021) medical students. We also found that gender, marital status, smoking status, sleep quality, IA, and religiosity are significantly associated with mental health problems.

This study found a higher prevalence of mental health problems among female students than male students, which is consistent with previous studies (Kerebih et al., 2017; Rtbey et al., 2022). The emotional traits of the way they respond to stress, domestic abuse, a sense of helplessness, and hormonal changes during menstruation might all be contributing factors to the increased frequency of mental discomfort among female students (Rtbey et al., 2022). The current study found that mental health problems among unmarried students were higher than those among married students, similar to previous studies (Kerebih et al., 2017). A possible reason why married medical students experience fewer mental health problems is that managing both academic and marital responsibilities requires effective time management, a positive attitude, and strong personal discipline (Dag-um et al., 2024).

This study found that internet addiction and poor sleep quality were significantly associated with poor mental health. A previous study conducted on medical students in Bangladesh revealed that sleep quality is significantly associated with mental health issues such as anxiety, stress, and depression (Muyeed et al., 2026). These findings were obvious because poor sleep quality increases internet addiction and decreases life satisfaction, which increases the probability of poor mental health (Sayeed et al., 2021; Scott et al., 2021). Additionally, the prevalence of poor mental health was higher among cigarette smokers than non-smokers (Kerebih et al., 2017). Medical students are frequently exposed to high academic demands and prolonged study hours, which may contribute to irregular sleep patterns and, subsequently, adverse

mental health outcomes (Wilkinson, 2023; Willcock et al., 2004).

In the current study, we found that religiosity was linked with the mental health status of medical students. Religious students suffer lower mental health problems, which may be due to the fact that medical students often experience stress due to academic pressure, social challenges, and uncertainty about their future, and religiosity can serve as a buffer against such psychological distress (Yusoff et al., 2010). Engaging in religious practices helps reduce mental health problems by promoting positive emotions and providing social support (Malinakova et al., 2024). Religiosity acts as a protective factor for medical students' mental health by promoting emotional resilience, self-regulation, and community support.

Another reason was that religious individuals were happier than non-religious individuals (Mondal et al., 2024), and it was known that happiness is associated with better mental health (Almadani & Alwesmi, 2023). However, in Bangladesh, religiosity may be associated with social acceptance, and positive feedback from friends and other people living in the society may be linked to better mental health outcomes.

This study has several strengths. To the best of our knowledge, it was the first study in Bangladesh to establish an association between religiosity and mental health. This study employed a robust sampling technique that minimizes sampling selection bias. We gathered data from multiple medical colleges, enhancing the generalizability of the findings to medical colleges throughout Bangladesh. The use of adapted and reliable psychometric tools increased the trustworthiness of the study's results. Collecting offline data reduces the likelihood of self-selection and response bias among participants. However, this study also has some limitations. Given the cross-sectional nature of the study, it is not possible to establish a cause-and-effect relationship between mental health and religiosity, internet addiction, and sleep quality. The reliance on questionnaires may introduce self-reporting bias due to social desirability. Another limitation of this study was that religiosity was assessed with a single question providing only two self-report options, which limited both conceptual depth and measurement accuracy. This binary response limits variability and may cause misclassification, potentially obscuring the true association with mental health issues. Nonetheless, future research should employ culturally valid, multidimensional instruments to assess both religious belief and practice, and should be conducted as a longitudinal study to examine the mediating and moderating mechanisms of sleep quality and religiosity, thereby enhancing understanding of the intricate relationship between explanatory variables and mental health.

7. Practical implications

The study's findings have several practical implications for enhancing the mental health of medical students, given the observed associations between poor sleep quality, internet addiction, and mental health issues. Suggested interventions include establishing structured sleep hygiene programs, launching awareness campaigns about the dangers of excessive internet use, and encouraging healthy digital habits. Early screening efforts could be introduced to identify students at risk of poor sleep, problematic internet use, or mental health problems. Providing mental health support services within academic institutions, such as psychotherapy or stress management programs, might help mitigate these risks. The identified association between religiosity and mental health suggests that culturally sensitive and contextually suitable coping strategies could benefit the well-being of medical students. Although these relationships cannot be interpreted causally, they highlight the potential importance of incorporating culturally sensitive techniques into students' support programs.

8. Conclusions

More than half of the medical students experienced poor mental health, highlighting a significant concern among them. Female and

unmarried students were more likely to develop mental health issues. Additionally, poor sleep quality and internet addiction are further related to worsening mental health problems. Furthermore, religiosity is associated with better mental health, suggesting a potential protective role. These findings emphasize the importance of holistic and inclusive approaches to developing mental health strategies that address both lifestyle and psychosocial factors among medical students.

CRediT authorship contribution statement

Md Abdul Hannan Mondal: Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nitai Kumar Saha:** Writing – review & editing, Writing – original draft, Supervision, Formal analysis, Data curation, Conceptualization. **Moy-nul Haque:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Masom Mia:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation.

IRB/ethical approval statement

The research was approved by the Ethical Review Committee, Rajshahi Medical College, Rajshahi-6000, Bangladesh (Ref. RMC/ERC/2024/158).

Data availability statement

We confirm that the data supporting the findings of this study will be shared upon reasonable request.

Statement for studies in humans or animals

The ethical approval was taken from the **Ethical Review Committee, Rajshahi Medical College, Rajshahi-6000, Bangladesh (Ref. RMC/ERC/2024/158)**. We also consider the guidelines provided by the 1964 Helsinki Declaration. We asked the participants for their consent to participate in this study. If they agreed, then we provided questionnaires. Then, we discussed the study's purpose and relevant questionnaires before collecting the final data. We did not provide any financial aid to participate in this study. All information was collected anonymously to consider the privacy and confidentiality of the participants.

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Declaration of competing interest

The authors have declared no conflict of interest.

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